Example 15.4

Transfer function

$$X_t = \sum_{j=-\infty}^{\infty} b_j Y_{t-j} \qquad b_j = \frac{1}{2m+1} \quad j \in \{-m, \dots, m\}$$

$$b_j = 0 \qquad \text{otherwise}$$

$$h(\omega) = \sum_{j=-m}^{m} \frac{1}{2m+1} e^{-i\omega j}$$

$$= \frac{1}{2m+1} \sum_{j=-m}^{m} e^{-i\omega j}$$

$$= \frac{1}{2m+1} \left[1 + \sum_{j=1}^{m} e^{-i\omega j} + \sum_{j=1}^{m} e^{i\omega j} \right]$$

$$= \frac{1}{2m+1} \left[1 + \underbrace{\sum_{j=1}^{m} (e^{-i\omega})^j}_{\text{GP } m \text{ terms}} + \underbrace{\sum_{j=1}^{m} (e^{i\omega})^j}_{\text{GP } m \text{ terms}} \right]$$

$$\left[S_n = \frac{a(1-r^n)}{1-r} \qquad \begin{array}{l} a = \text{First term} \\ r = \text{Common ratio} \end{array} \right]$$

$$= \frac{1}{2m+1} \left[1 + \frac{e^{-i\omega}(1 - e^{-im\omega})}{1 - e^{-i\omega}} + \frac{e^{i\omega}(1 - e^{im\omega})}{1 - e^{i\omega}} \right]$$

$$= \frac{1}{2m+1}\left[\frac{(1-e^{iw})(1-e^{-iw})}{(1-e^{iw})(1-e^{-iw})}\right.$$

$$+ \frac{e^{-iw}(1-e^{-imw})(1-e^{iw})}{(1-e^{iw})(1-e^{-iw})}$$

$$\left. + \frac{e^{iw}(1-e^{imw})(1-e^{-iw})}{(1-e^{iw})(1-e^{-iw})}\right]$$

$$= \frac{1}{(2m+1)(2-2\cos w)}\left[2 - 2\cos w\right.$$

$$+ e^{-iw}(1-e^{-imw} - e^{iw} + e^{-i(m-1)w})$$

$$\left. + e^{iw}(1-e^{imw} - e^{-iw} + e^{i(m-1)w})\right]$$

$$= \frac{1}{(2m+1)(2-2\cos w)}\left[2\cos(mw) - 2\cos(m+1)w\right]$$

$$= \frac{1}{(2m+1)(1-\cos w)}\left[\cos(mw) - \cos(m+1)w\right]$$

「 $\cos A - \cos B = -2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)$ 」